

Lecture 7

Cumulative Distributions

Review

1 Why do we square the deviations before averaging them?

Because the deviations always sum to zero – the positives cancel the negatives, so their average is 0 for every dataset.

2 The variance came out in squared units. What fixes that?

Taking the square root. The standard deviation is back on the scale of the data.

3 Why divide the sum of squares by $n - 1$ rather than n ?

The deviations must sum to zero, so once you know $n - 1$ of them the last is forced. Only $n - 1$ are free to vary.

4 Where do the two fences sit, and what do they decide?

At $Q1 - 1.5(IQR)$ and $Q3 + 1.5(IQR)$. Anything outside them is an outlier, and the whiskers stop before them.

The data: sugar in twenty breakfast cereals

- X = grams of sugar per serving, for twenty cereals on a US supermarket shelf
- $n = 20$, so n is even - keep that in mind, it changes how we read one thing later

Cereal	Sugar (g)	Cereal	Sugar (g)
Fiber One	0	Cinnamon Toast Crunch	10
Cheerios	1	Frosted Mini Wheats	11
Corn Flakes	3	Honeycomb	11
Rice Krispies	3	Captain Crunch	12
Special K	4	Frosted Flakes	12
Wheaties	4	Apple Jacks	14
All Bran	5	Froot Loops	14
Life	6	Honey Smacks	15
Honey Bunches of Oats	7	Crackling Oat Bran	16
Honey Nut Cheerios	9	Raisin Bran	18

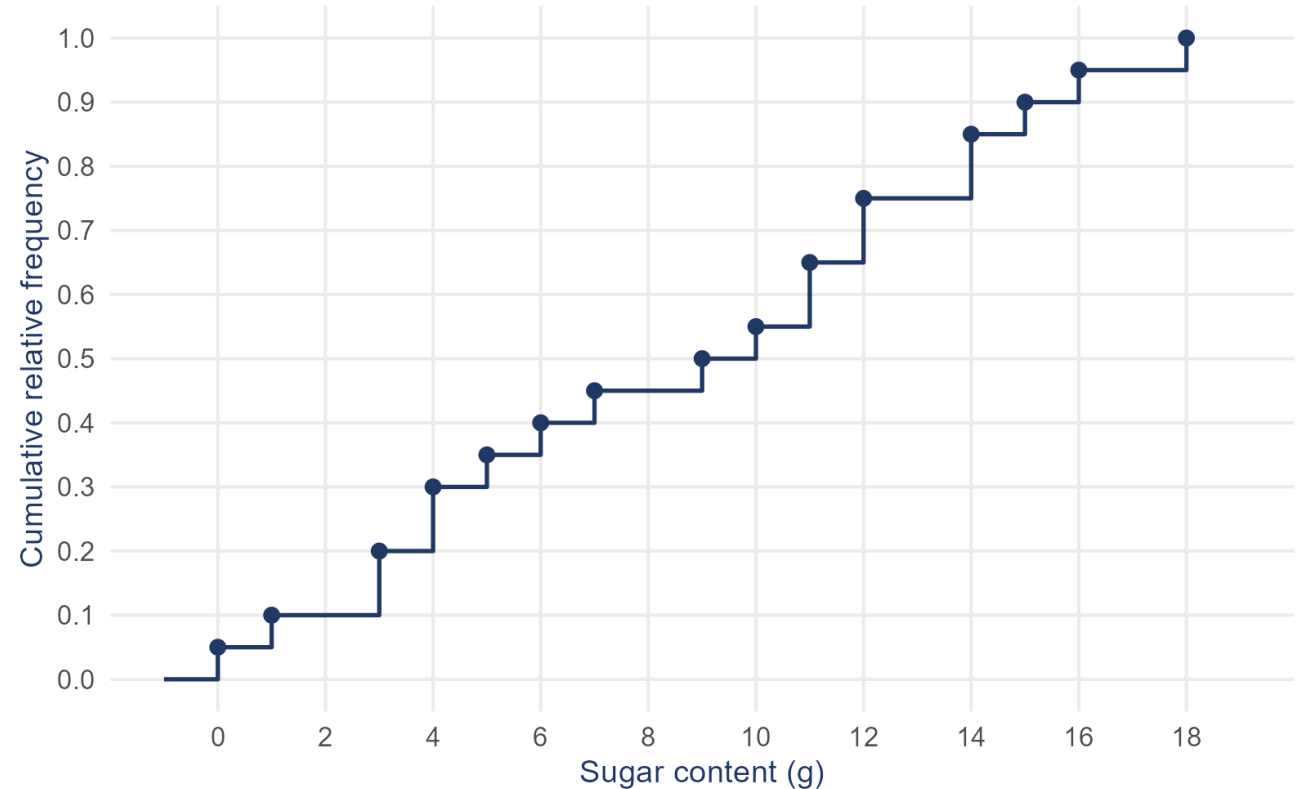
Cumulative distributions

- A **cumulative distribution** shows the relationship between the value of a variable and the cumulative relative frequency
- $\text{crf}(x)$ answers one question: what fraction of the data falls at or below x ?
 - It only ever climbs - it can never come back down
 - It starts at 0 and finishes at 1
- For discrete data we draw it as a **step function**: flat between observed values, jumping at each one
 - The height of a jump is that value's relative frequency

x	f(x)	rf(x)	crf(x)
0	1	0.05	0.05
1	1	0.05	0.10
3	2	0.10	0.20
4	2	0.10	0.30 ← crf passes 0.25
5	1	0.05	0.35
6	1	0.05	0.40
7	1	0.05	0.45
9	1	0.05	0.50 ← crf = 0.50 exactly
10	1	0.05	0.55
11	2	0.10	0.65
12	2	0.10	0.75 ← crf = 0.75 exactly
14	2	0.10	0.85
15	1	0.05	0.90
16	1	0.05	0.95
18	1	0.05	1.00

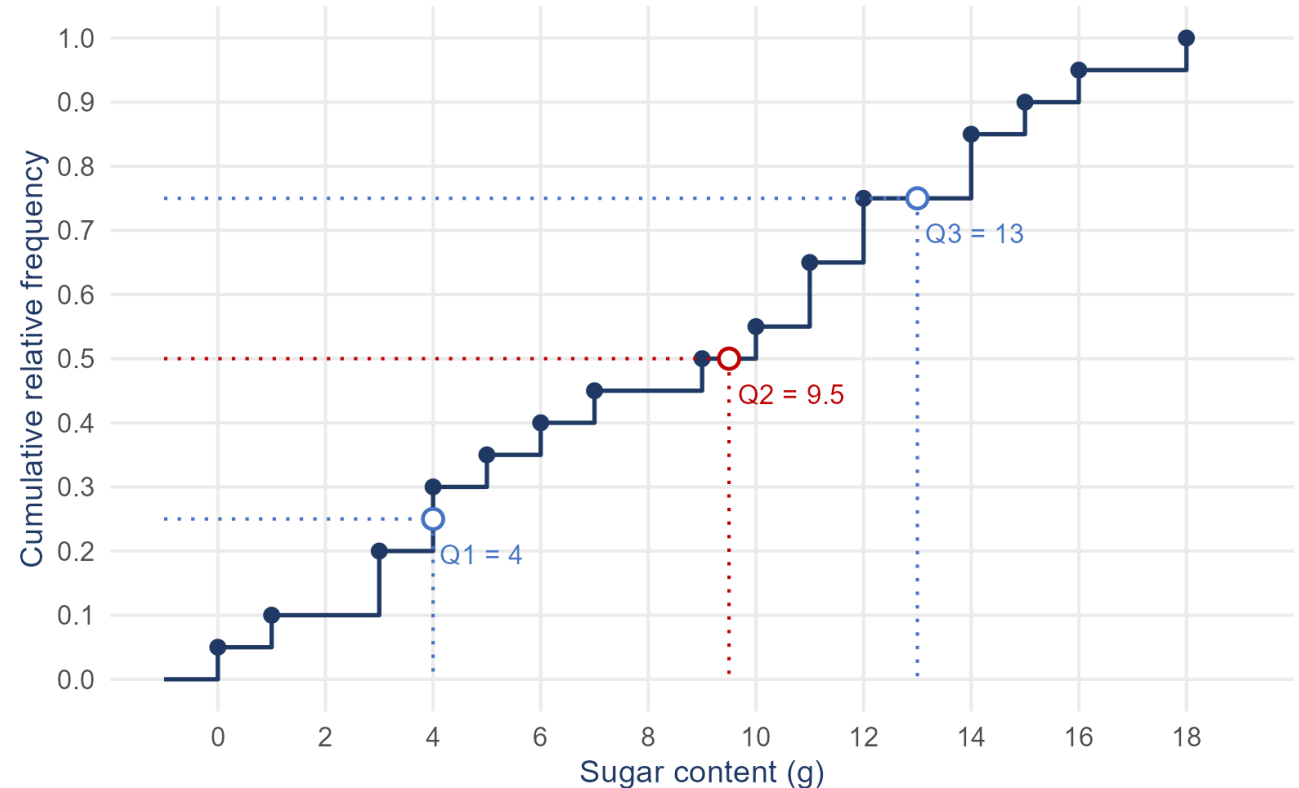
From the table to the plot

- Put the values of the variable on the x-axis and $\text{crf}(x)$ on the y-axis
- Plot a point at each observed value
- Hold that height flat until the next observed value, then jump
- The flat runs are not decoration - they are what we read a percentile off



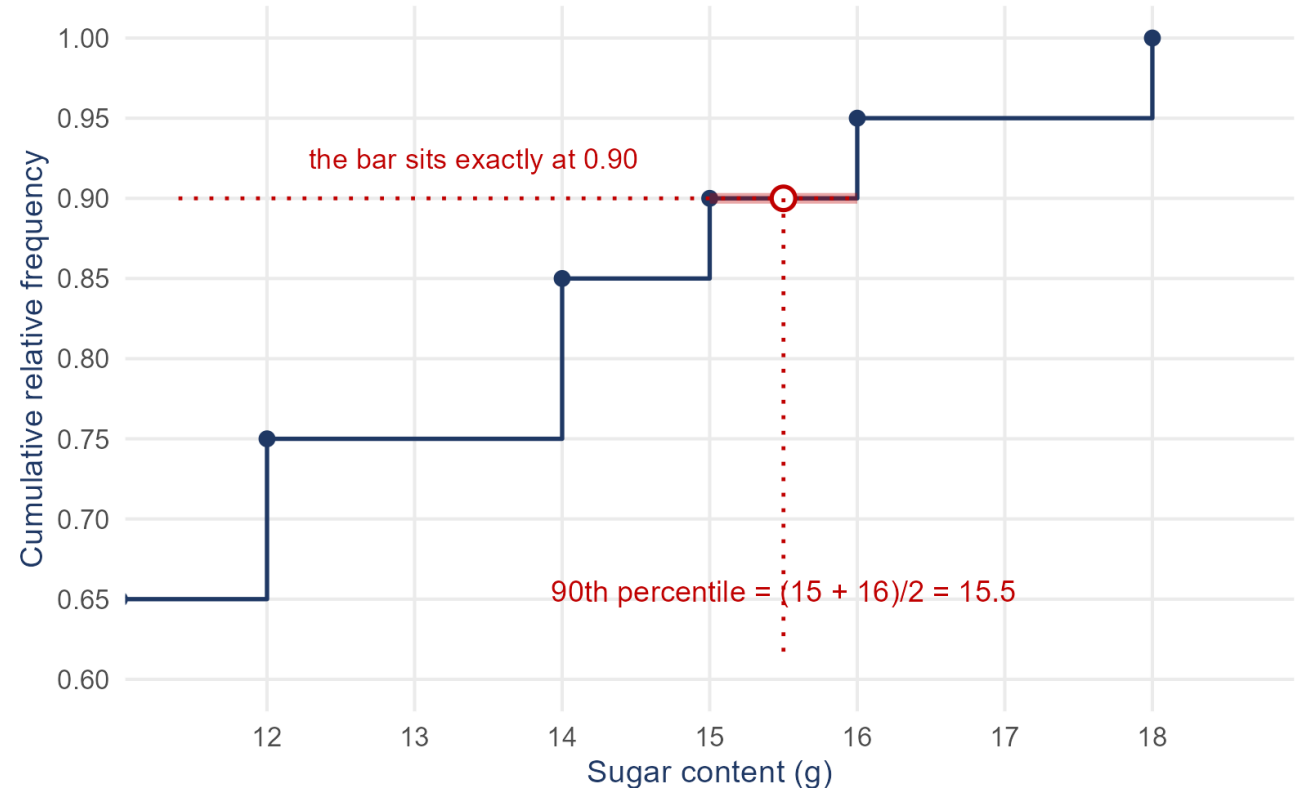
Reading a percentile off the plot

- To find the pth percentile:
 - go up the y-axis to $p/100$
 - read across to the curve
 - drop down to the x-axis
- Q1, Q2 and Q3 are just the percentiles at 0.25, 0.50 and 0.75
 - **Q1 = 4** **Q2 = 9.5** **Q3 = 13**
 - **IQR = 13 - 4 = 9**



When the bar sits exactly on the percentile

- Reading across at p usually crosses a jump, and the answer is that value
- But sometimes the line lands ON a flat run instead
 - then the p th percentile is the **midpoint** of that bar
- This happens when n is even, which is why $n = 20$ mattered
 - crf reaches 0.90 exactly at $x = 15$, and the next value is 16
 - so the 90th percentile is $(15 + 16)/2 = 15.5$, not 15
- Same story at 0.50 and 0.75, which is why $Q2 = 9.5$ and $Q3 = 13$



Going the other way: what percentile is this value at?

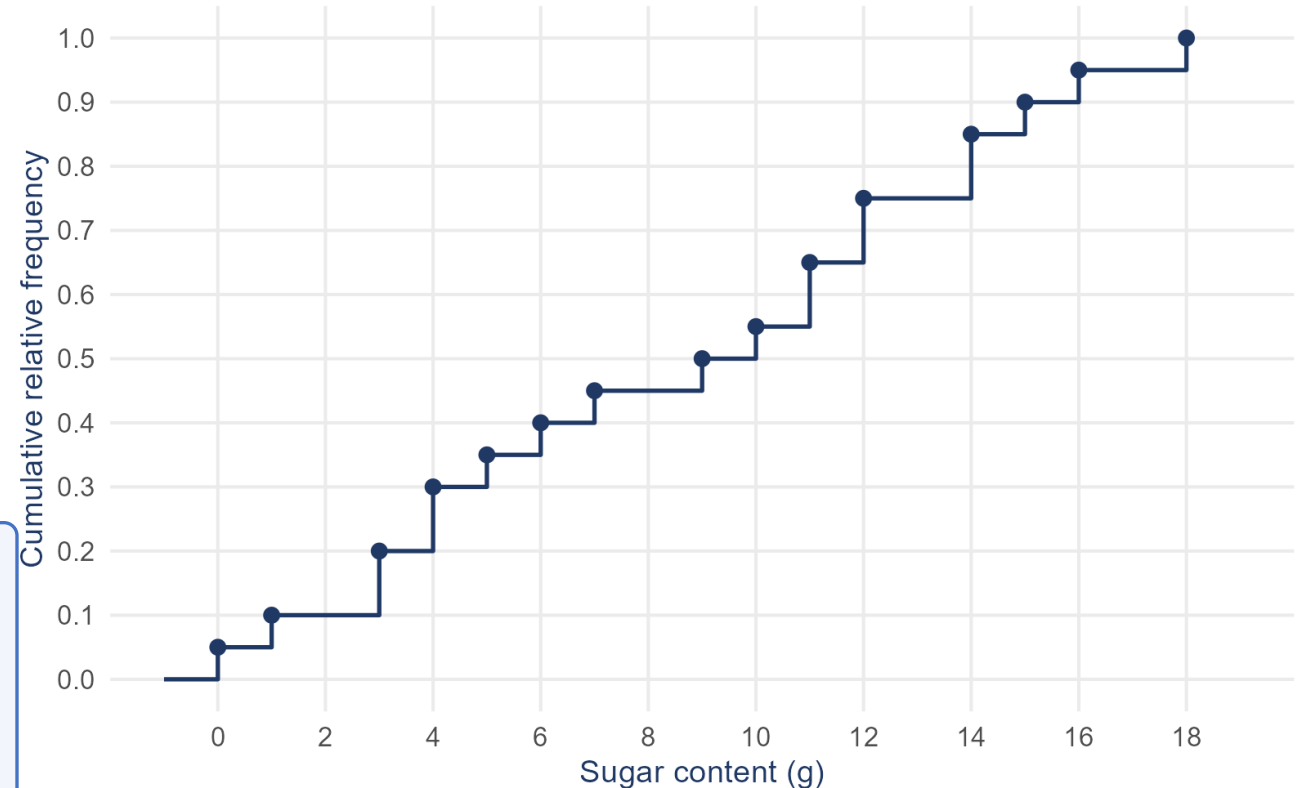
- The same plot answers the reverse question. Start on the x-axis instead:
 - go along to the value
 - up to the curve
 - across to the y-axis, and read crf
- Your turn: Honeycomb has 11 g of sugar. What percentile is it at?
- And what about All Bran, at 5 g?

Working:

11 g $\rightarrow$ crf = 0.65 $\rightarrow$ 65th percentile

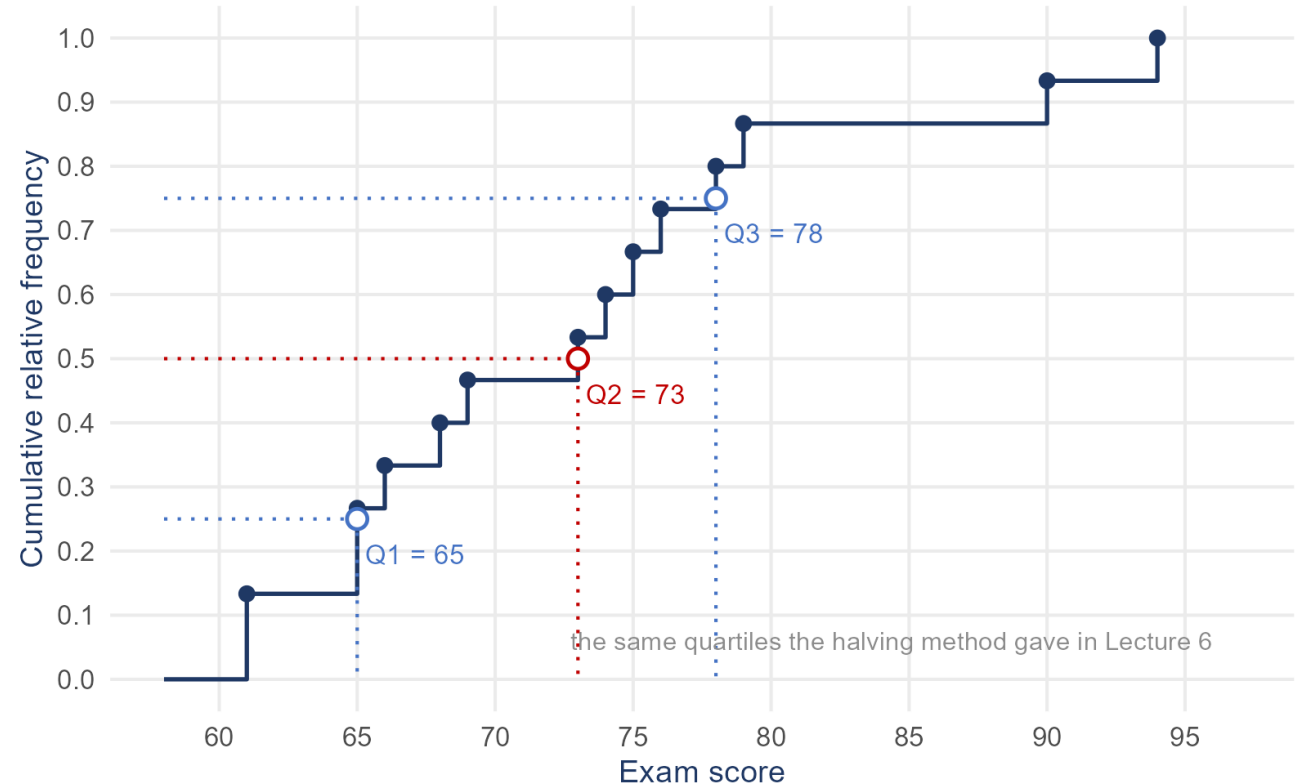
5 g $\rightarrow$ crf = 0.35 $\rightarrow$ 35th percentile

65% of cereals have at most as much sugar as Honeycomb.



Practice: quartiles from a cumulative distribution

- This is the cumulative distribution of the 15 exam scores from Wednesday
- Read Q1, Q2 and Q3 off the plot
- $n = 15$ is odd this time. Does the midpoint rule come into it?
- Then check your answers against the ones you got by halving the ordered data



Two pictures of the same five numbers

- The step plot and the boxplot are not competing summaries - the boxplot is five readings off the step plot
 - the minimum is where the curve leaves 0, the maximum is where it reaches 1
 - Q1, Q2 and Q3 are the readings at 0.25, 0.50 and 0.75
- What each one is good for:
 - the step plot gives you ANY percentile, not just the quartiles
 - the boxplot is far easier to compare across groups, and it flags outliers
 - the histogram is still the only one of the three that shows peaks and gaps

Descriptive statistics: the whole toolkit

- Every question we have asked about a single variable falls into four buckets
 - **SHAPE** - symmetric, skewed, how many peaks, gaps and outliers (histogram, dot plot, stem and leaf, boxplot)
 - **CENTER** - mean, median, mode
 - **SPREAD** - range, IQR, variance, standard deviation
 - **POSITION** - percentiles and quartiles (ordered data, or the cumulative distribution)
- Resistant to outliers: the median, the mode, the IQR
- Not resistant: the mean, the range, the variance and the standard deviation
- Shape decides which pair you report - mean with s, or median with IQR

End of the material for Exam 1

- Everything on Exam 1 has now been covered. Monday starts new material that is NOT on it.
- Wednesday September 16 - in-class review
- Friday September 18 - Exam 1, in class
 - delivered through Canvas, so bring a charged laptop
 - a calculator is allowed, as long as it cannot communicate wirelessly
 - notes are not permitted; the formula sheet is printed on the study guide
- The study guide and the review problems are on the course website, under Exam Study Guides
- **Homework 1** is due Sunday at 11:59 PM